Neural-Grounded Computational Decision Phenotypes: Honest-by-Construction Prediction of Choice, with Provable Abstention

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Abstract

Predicting how a person will decide is usually treated as a black-box classification problem, which gives uncalibrated point predictions and no principled way to say "I do not know". We take a different route grounded in the neuroscience of choice. Following the Affect-Integration-Motivation account, we describe a decision-maker by a small set of neuroeconomic parameters and split the choice process into an affective component that generalizes across people and an integrative component that individuates. We estimate this with a drift-diffusion likelihood, and wrap every prediction in distribution-free conformal coverage, a strictly-proper score, and a minimum-detectable-effect abstention gate, so the model abstains rather than guesses when the data cannot support a prediction. We validate the estimator, honesty layer, and pooling on real human choices from two mixed-gambles fMRI datasets (124 subjects); the phenotype has low-dimensional structure, predicts held-out choice at AUC 0.96, individuates by cross-run fingerprinting, and transfers across labs. We ground the affective channel in real neural data across two modalities (fMRI loss channel; EEG reward positivity) and, where a direct within-subject neural-to-behavior link is underpowered, the model abstains and a correlation-transitivity argument shows the grounding does not require it. We report simulation results as positive/negative controls and real-data results as measured effects throughout.

Preprint. Every number in this manuscript traces to a committed analysis script and a result file: simulation results are positive/negative controls on the method, and real-data results are measured on public human datasets (NARPS ds001734; Tom 2007 ds000005; EEG ds003458). Code and data-processing scripts are available on request.

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1. Introduction

Predicting a person's decisions is high-stakes, yet most methods are silent about their own uncertainty. The neuroforecasting literature (Genevsky & Knutson 2015; Genevsky, Yoon & Knutson 2017; Tong et al. 2020; Stallen et al. 2021; Genevsky, Tong & Knutson 2025) showed that brain-derived *affective* signal can forecast aggregate, out-of-sample choice better than behavior, and generalizes even when the scanned sample is unrepresentative. Those results, however, are population-to-population, at small sample sizes, and have never been unified into a single identifiable estimator or equipped with distribution-free guarantees.

We contribute a method that is (i) neurally grounded in the Affect-Integration-Motivation account, (ii) honest by construction (conformal coverage, proper scoring, and an abstention gate), and

(iii) explicit about the one thing prior work leaves implicit: *when a prediction is not licensed at all.* Throughout, we do not claim to measure any individual's brain: the neural grounding is a population property, and claims about a specific decision-maker are about their revealed decision policy, bounded by named confounds.

1.1 Contributions

The neuroforecasting phenomena we build on are established; our contributions are methodological and several are new empirical findings, not re-analyses:

1. **Abstention as a first-class, provable output.** To our knowledge this is the first neural-grounded behavioral-prediction model that *withholds* a prediction when the data cannot support one, via a distribution-free coverage guarantee, a strictly-proper score, and a minimum-detectable-effect gate.
2. **The attribution-scale ladder**, a new conceptual axis that turns the "who owns a decision" confound into a measurable variable and yields a dose-response for *where* individual-level neural inference is licensed.
3. **Grounding by triangulation, made rigorous.** Using the positive-semidefinite correlation-transitivity bound, we show the direct within-subject neural-to-behavior link is *constrained by the multi-modal chain and not required* — a principled criterion for when convergent grounding suffices without a direct causal link.
4. **A cautionary finding about a common measure.** On real fMRI, the widely-used nucleus-accumbens loss/gain *ratio* index of neural loss aversion is unstable and sign-unstable; it was the only nominally significant correlate while the stable channel measures were near zero, and our gate abstains exactly where naive null-hypothesis testing would report an effect.
5. **An identifiability result:** from choice data alone the scale-free affective ratios are recoverable but the decision temperature is not (it requires reaction times).
6. **Two phenotype results on real data:** the decision phenotype transfers across labs (out-of-dataset choice prediction) and individuates by cross-run fingerprinting, establishing it as a stable, transferable individual trait.

2. Related work

- **Neuroforecasting / AIM:** Knutson (2008) anticipatory affect; Samanez-Larkin & Knutson (2015) AIM; Knutson & Genevsky (2018) affective-vs-integrative transfer; Tong (2020), Stallen (2021), Smith/Montague (2014), Falk (2012/15). Fernandes (2022) gives AIM an EEG signature, which licenses our EEG arm.
- **Forecast evaluation:** Gneiting & Raftery (2007) strictly-proper scoring rules.
- **Distribution-free uncertainty:** Angelopoulos & Bates (2021) conformal prediction.
- **Choice process models:** drift-diffusion / sequential sampling; resource-rational analysis (Lieder & Griffiths 2020) as *interpretation* of the parameters, not a predictive claim.
- **Inverse inference:** Bayesian inverse planning and theory of mind as inverse RL (Baker, Saxe & Tenenbaum 2009; Jara-Ettinger); computational-psychiatry generative modeling and its identifiability discipline (Huys, Montague); neural latent-state inference (Schiereck et al. 2025; Blanco-Pozo et al. 2024). Prospect theory in foreign-policy decision-making (McDermott; Levy), where leaders' latent risk postures are explicitly *inferred* from choices, motivates the attribution-scale ladder.
- **Real-data paradigm:** the mixed-gambles reward task (Tom, Fox, Trepel & Poldrack 2007) and its large public replication dataset NARPS (Botvinik-Nezer et al. 2019 Sci Data; 2020

Nature) supply real human loss-aversion choices and reward-anticipation fMRI.

- **Gap we fill:** no prior work unifies AIM decomposition + a choice-process likelihood + distribution-free coverage + an identifiability-linked abstention gate in one estimator, nor evaluates abstention as a safety property on confounded real-world decision records.

3. Method

3.1 The decision manifold

theta = (rho_gain, lambda_loss, kappa_risk, delta_time, omega_threat, tau_consistency). An event is a context vector c (stakes, gain/loss frame, ambiguity, threat, time pressure, reference point).

3.2 AIM-structured valuation and DDM likelihood

V(a | c, theta) = rho_gain E[gain] - lambda_loss E[loss] - kappa_risk sigma - omega_threat ambiguity + delta_time delay. The value difference drives a drift-diffusion process; boundary set by caution. Affective channel (rho, lambda, omega) shares a hierarchical population prior; integrative channel (kappa weighting, tau/threshold) is per-agent. This is the operationalization of Knutson & Genevsky's affective-vs-integrative split.

3.3 Estimation

Hierarchical (empirical-Bayes) partial pooling shrinks sparse individuals toward the population mean so underpowering shows up as estimate variance, not overconfidence. From choice data the identifiable coordinates are the scale-free affective ratios (loss aversion, threat, risk, discount relative to gain sensitivity); the decision temperature tau requires reaction times (the full DDM), which is a deployment-step addition.

3.4 Honesty layer (C2)

Strictly-proper scoring (log/Brier/CRPS); split-conformal prediction sets at nominal coverage; an MDES gate that abstains when the identifiable-event count puts the minimum detectable effect above the claimed effect. Abstention is a first-class output.

3.5 Neural grounding

Affective-channel priors and the valuation form are grounded in real recordings across two neural modalities: NAcc/insula betas (fMRI, NARPS) and the EEG Reward Positivity (open dataset ds003458; DEAP, the licence-gated alternative, is not required). No individual target is scanned.

3.6 Inverse inference: from a decision to the latent state that best explains it

The forward model predicts a decision from theta and c. The **inverse** problem runs it backward: given an observed decision D and its context c, infer the posterior p(z | D, c) over the latent decision-state z, where z is the affective posture (the gain-approach vs threat-avoidance balance and the risk posture) that best explains D under the forward AIM-DDM model. This is Bayesian inverse planning (Baker, Saxe & Tenenbaum 2009; theory of mind as inverse reinforcement learning) applied to a neurally-structured generative model, and it is a computational-phenotyping inference of the kind that computational psychiatry performs with explicit attention to parameter identifiability (Huys, Montague and others). Because the forward map is calibrated on scanned populations, z carries a **neural interpretation** (a gain-approach posture is the computational analog of elevated NAcc-type gain-anticipation relative to insula-type threat-anticipation). We report that interpretation as a hypothesis attached to a latent, with its posterior width, and we never state it as measured neural firing. The honest object is "decision D is consistent with an affective posture tilted toward gain-approach, posterior width W", not "region R fired in person P".

3.7 The attribution-scale ladder: where does inverse inference survive?

Neuroforecasting is validated population-to-population; the aggregate affective signal is exactly the component that generalizes. The attribution confound (was a decision the individual's or the institution's) is therefore not only a nuisance, it is an axis. We evaluate inverse identifiability at nested attribution scopes, from most aggregate to most individual:

- S1 executive branch including all agencies; S2 White House and the Pentagon; S3 the White House; S4 President and cabinet; S5 President and key advisors; S6 the President alone.

The prediction from AIM and the neuroforecasting results is a **dose-response**: inverse identifiability (posterior sharpness, and out-of-sample skill of the recovered latent) is highest at aggregate scopes, where the population-generalizable affective component dominates, and degrades as scope narrows toward the individual, where the idiosyncratic integrative component and the attribution confound take over. The abstention gate (Section 3.4) should engage increasingly as scope narrows. The empirical question is the title of the ladder: how far toward S6 can inference survive before the gate must abstain. A sharp individual-scope posterior would be a red flag (analyst degrees of freedom), not a triumph. **Subject and decision selection:** US presidents with dense public policy records, and within each, decisions with short response latency (rapid crisis reactions), because the anticipatory- affect channel that neuroforecasting relies on operates on that fast timescale, so fast reactive decisions are where the affective (generalizable) component is most applicable and slow deliberative ones are dominated by the integrative (individuating) component.

4. Experiments

Experiments E1-E4, E6, E7 are run on controlled data with known ground truth (done). E5 and the real-data instances of E3/E6 are the deployment step (Section 7, pending data).

- **E1 (C1) Parameter recovery.** Simulate agents at known theta; fit; correlate recovered vs true affective *ratios* across agents; 20 seeds; report the recovery curve over decisions-per-agent with CIs. Pass: affective ratios reach r >= 0.6 within an achievable per-agent decision count.
- **E2 (C2) Coverage, calibration, abstention.** Empirical split-conformal coverage vs nominal on a heteroscedastic problem; expected calibration error; the MDES gate on under- vs adequately-powered effects. Pass: coverage within 0.03 of nominal, low ECE, gate abstains when underpowered.
- **E3 (C3) Affective-vs-behavioral transfer, controls.** Within the estimator, forecast a per-stimulus market outcome from an aggregate affective (brain) arm vs an aggregate behavioral arm vs a content-only baseline; stimulus-grouped CV; out-of-sample R2 + CRPS + permutation null. Positive control (market tracks affect) and negative control (market is noise). Pass: brain > behavior on the positive control, null on the negative. Real NARPS (fMRI) + an open EEG dataset (ds003458), plus the content-blind 2-factor specificity ablation, are the deployment.

- **E4 (C4) Pooling buys power.** Unpooled vs empirical-Bayes pooled recovery r at low decisions-per-agent. Pass: the bootstrap CI on the improvement excludes zero.
- **E5 (C5) Calibrated abstention on a real decision corpus (deployment, pending).** Public-record decision corpus (two blind coders + Cohen's kappa; preregistered temporal split); conformal predictions where identifiable; abstention where confounds break identifiability; calibration on the non-abstained subset vs base rate.
- **E6 (C5, C6) Inverse inference + attribution-scale ladder.** Recover the latent affective posture across attribution scopes S1..S6; measure identifiability (recovery r) and abstention rate per scope. Pass: the predicted dose-response (identifiability falls, abstention rises, S1->S6). The latent is an affective posture with a neural interpretation, never measured firing.
- **E7 (extension) States as recoverable manifold displacements.** An entity decides in a baseline and a threat-elevated state (threat coordinate shifted by a known amount); recover the per-condition posture and test whether the recovered shift tracks the true shift, with a zero-shift negative control. Pass: shift detected on the positive control, null on the negative.

5. Results

The results below are validations of the estimator and the honesty layer on controlled data: parameter recovery against known ground truth (E1), the coverage/calibration/abstention guarantees of the honesty layer (E2), pooling's effect on low-data recovery (E4), and positive/negative controls on the transfer pipeline's operating characteristics (E3). They establish that the instrument measures what it claims and does not manufacture effects; the empirical neuroforecasting claim itself is evaluated on real fMRI + EEG (NARPS; ds003458), the deployment step (Section 7).

E1 (C1) Identifiability, and its data requirement (Fig 3, Table 2). Recovery of the affective phenotype ratios rises with decisions per agent (20 seeds, 60 agents). Raw logistic coefficients recover at r = 0.53 to 0.88 across 200 to 2000 decisions/agent, confirming the model is well specified. The scale-free affective ratios reach the r >= 0.6 criterion by roughly 1000 to 2000 decisions/agent (threat r = 0.60 at 1000 and 0.73 at 2000; loss aversion r = 0.69 at 2000) and are underdetermined at 200. This data requirement is the quantitative motivation for pooling (C4) and for abstention on sparse records (C5). The decision temperature tau is not identifiable from choice alone and is recovered from reaction times with the full DDM (future work, Section 7).

E2 (C2) Honest-by-construction (positive control, `results/e2_honesty_selfcheck.json`). Split-conformal intervals achieve nominal coverage (empirical 0.90 at nominal 0.90 on a heteroscedastic problem, within 0.03); a well-specified probability model has expected calibration error 0.018; the MDES gate abstains on an underpowered effect (r = 0.15 at n = 50) and reports a powered one (r = 0.30 at n = 500). The layer delivers coverage, calibration, and abstention as designed.

E4 (C4) Pooling buys power (Fig 6). Empirical-Bayes partial pooling lifts low-data recovery: at 150 decisions/agent, recovery r improves for every ratio with a bootstrap CI on the improvement excluding zero (loss aversion 0.10 to 0.30, threat 0.15 to 0.27, risk 0.09 to 0.44, discount 0.15 to 0.44; 20 seeds). Borrowing strength across the population reduces per-agent estimate variance, the mechanism behind a lower effective MDES.

E3 (C3) Transfer pipeline operating characteristics (Fig 5). On the positive control, where the aggregate market outcome tracks the affective latent by construction, the aggregate affective (brain) arm forecasts held-out stimuli better than the aggregate behavioral arm (out-of-sample R2 0.845 vs 0.731; brain-minus-behavior Delta R2 = +0.114, 95% CI [+0.092, +0.137]; stimulus-grouped permutation p = 0.003). On the negative control, where the market is independent noise, no arm exceeds chance and the permutation null is not rejected (brain R2 = -0.05, permutation p = 0.90, brain-minus-behavior CI includes 0). The pipeline detects the dissociation only when it exists. This is a synthetic positive/negative control on operating characteristics, not the empirical claim.

E6 (C5, C6) Inverse inference is scale-dependent (Fig 7). Running the estimator in reverse to recover the latent affective posture, evaluated across nested attribution scopes S1 (executive plus agencies, 150 contributors) to S6 (the individual, one contributor), yields the predicted dose-response (50 entities, 8 seeds). Identifiability of the posture falls monotonically from r = 0.35 at the aggregate scope to r = 0.11 at the individual scope, while the abstention rate rises from 0.16 to 0.86. On the subset the gate does not abstain, recovery stays higher than on the full set at every scope (for example 0.57 vs 0.35 at S1), so abstention keeps what is reported more reliable. The individual scope is where inference is mostly not licensed: the framework abstains on 86 percent of entities there, exactly the behavior the attribution confound predicts and the safety property C5 asserts. The latent is a computational affective posture with a neural interpretation, never a claim of measured firing; this simulation establishes the mechanism, and the real US-president decision corpus (two blind coders and Cohen's kappa) is the deployment.

E7 (extension) States are recoverable displacements. A threat-elevated state is recovered as a detected, direction-correct shift in the threat coordinate (mean recovered shift +0.73, 95% CI excluding zero, tracking the true shift at r = 0.35 across 60 entities, 10 seeds), while a zero-shift negative control recovers no shift (mean +0.002, CI includes zero). The same estimator that fits a stable phenotype also detects its transient displacement under a state.

5.1 Real data: the phenotype on NARPS ds001734 (Fig 8)

We ran the estimator on real human choices from the NARPS mixed-gambles task (Botvinik-Nezer et al. 2019; the Tom, Fox, Trepel & Poldrack 2007 paradigm): 108 subjects, 27,454 accept/reject decisions over 50/50 gain-vs-loss gambles, with reaction times (OpenNeuro ds001734, CC0, events only).

- **RN1 (C1 on real data).** Per-subject loss aversion lambda = |b_loss|/|b_gain| recovered from real choices, split by the study's range manipulation: the equalIndifference group (asymmetric gain/loss range, as in Tom 2007) has median lambda = 1.45, while the equalRange group (symmetric range) has median lambda = 1.00, the known range-adaptation of loss aversion. Within-subject held-out choice prediction: mean out-of-sample AUC = 0.958 (95% CI [0.951, 0.964]), Brier 0.061 vs a base-rate Brier of 0.233.
- **RN2 (C2 on real data).** Split-conformal coverage on real held-out choices is 1.00 (>= the 0.90 nominal, conservative because the label set is small and predictions are confident), expected calibration error 0.034; the identifiability gate abstains on 1% of subjects.
- **RN3 (C4 on real data).** Empirical-Bayes pooling reduces the loss-aversion SD only slightly (0.80 -> 0.78), because every NARPS subject has ~256 trials, the high-n regime where the synthetic E4 predicted pooling helps least. The controlled and real results agree on when pooling matters.
- **RN4 (DDM signature on real data).** Reaction time correlates negatively with net decision-value magnitude (median r = -0.30 across 107 subjects): larger value magnitude, faster choice, the evidence-accumulation signature the DDM predicts, motivating the reaction-time DDM for tau.

So C1, C2, and C4 hold on real human decisions, not only in simulation.

5.2 Real fMRI: affective grounding, with the framework gating its own neural arm (Fig 9)

We fit per-subject first-level GLMs with gain and loss parametric regressors on the fmriprep-preprocessed NARPS BOLD (MNI152, one run, 40 subjects) and extracted NAcc and anterior-insula betas (6mm spheres). The loss/threat side of the affective channel is now grounded on real data: **NAcc decreases to loss ($t = -2.12$, $p = 0.040$) and anterior insula increases to loss ($t = +2.32$, $p = 0.025$)**, both AIM-consistent and significant across subjects. NAcc gain-tracking is in the predicted positive direction but not yet significant ($t = +1.47$, $p = 0.15$), and insula does not track gain ($p = 0.63$, as expected). All three AIM direction predictions hold in sign.

The cross-subject correlation between neural and behavioral loss aversion strengthened sharply from $n = 12$ to $n = 40$ ($r = -0.01$ -> -0.40), consistent with a real effect emerging with power, but at $n = 40$ the minimum detectable correlation is 0.44, so $|r| = 0.40$ sits just under the identifiability bar and the honesty gate abstains. This is the gate behaving exactly as designed at the boundary: an r of 0.40 with MDES 0.44 is suggestive, not established, and is reported as withheld rather than claimed. (The neural loss-aversion ratio is also sign-unstable at small per-subject betas, a further reason to abstain.) The loss-channel grounding is a real, significant neural result; the individual neural-to-behavioral link needs a larger sample or multi-run per-subject estimates to cross the bar.

5.3 The phenotype through three lenses: structure, function, individuation (Fig 10)

A decision phenotype earns the name only if it has geometric structure, predicts behavior (function), and identifies the individual. We evaluate all three on real NARPS data and fold them into the claims.

- **Structure (the manifold has real geometry).** Across 108 subjects the affective coordinates are strongly coupled: gain sensitivity and loss sensitivity correlate at $r = -0.92$, and the six-feature phenotype has an effective dimensionality of 4.4 (participation ratio), below its feature count. The population phenotype occupies a low-dimensional structure, as the manifold model (C1) assumes, rather than filling the feature space.
- **Function (it predicts choice, and grounds in reward circuitry).** The phenotype forecasts real held-out choices at AUC 0.96 (Section 5.1), and its affective coordinates align in direction with NAcc gain-tracking and insula loss-tracking (Section 5.2, underpowered). This is the affective->choice function the estimator is built on (C1, C3).
- **Individuation (it is an individual trait).** Cross-run fingerprinting identifies a subject in held-out runs from their run-01 phenotype at 13% accuracy versus 0.9% chance (14x, permutation $p = 0.0005$, differential identifiability $I\_diff = 1.10$). The phenotype is stable and person-specific, which is exactly the "integrative component individuates" half of the AIM split the framework rests on, and the property the C5/C6 individual-scope analysis depends on.

Read together, the three lenses say the phenotype is a real, low-dimensional, behavior-predictive, individuating object on real human data, not a curve-fit. Each lens strengthens a specific claim: structure and function under-write C1/C3, individuation under-writes the AIM split and the C5/C6 individual-scope reasoning.

5.4 Cross-dataset generalization: the phenotype transfers across labs (Fig 11)

We pooled a second, independent mixed-gambles dataset, Tom, Fox, Trepel & Poldrack (2007) (ds000005, 16 subjects, the original loss-aversion study), with NARPS. Two results. First, replication: the Tom-2007 loss aversion median is lambda = 1.94, reproducing the classic value, while NARPS is 1.12 (diluted by its symmetric-range equalRange group); the distributions legitimately differ (KS $p = 0.004$) because of the range manipulation, not a failure of the estimator. Second, and more important, cross-dataset transfer: a valuation model fit on NARPS predicts Tom-2007 choices out-of-dataset at AUC 0.86, and a model fit on Tom-2007 predicts NARPS choices at AUC 0.89. The affective valuation the estimator recovers is not a per-dataset artifact; trained in one lab it forecasts choices collected in another, years apart. This is the strongest real form of the pooling and external-validity claims (C4, Gate 7): the phenotype generalizes across datasets.

5.5 External confirmation of the gain channel in independent reward maps (Fig 12)

Our $n = 40$ NARPS GLM left the NAcc gain channel non-significant ($p = 0.15$), the one AIM direction we could not establish internally from single-run data. We sampled NAcc and anterior insula in independent group-level reward maps on NeuroVault. In an expected-value monetary-incentive-delay map (104 subjects) NAcc shows positive expected-value activation (sphere mean +1.0), in a gain > no-gain map (46 subjects) NAcc is +1.6, and in a social-reward-anticipation map NAcc is +0.7. The gain channel that our own sample was underpowered to confirm is positive across three independent, larger reward-anticipation samples. This is group-level external confirmation of the affective grounding, not per-subject power; it strengthens the channel claim while the individual neural-to-behavioral correlation still awaits a dataset with per-subject fMRI and choices together.

5.6b EEG cross-modal grounding: the Reward Positivity (Fig 14)

To ground the affective channel in a second neural modality, we computed the Reward Positivity on an open EEG gambling dataset (ds003458, 23 subjects; DEAP is licence-gated and not required). The frontocentral ERP to win feedback is more positive than to loss feedback by +3.0 microvolts (paired $t = 6.58$, $p < 10^{-6}$, Cohen's $d = 1.37$): a large, well-established EEG signature of reward processing. So the neural -> construct half of the grounding chain now holds across fMRI (loss channel) and EEG (reward channel), in independent datasets and modalities.

5.6 Grounding by triangulation, and its statistical validation (Fig 13)

No single public dataset carries per-subject neural signal and per-subject behavior in a form that resolves the individual neural-to-behavioral link. Rather than force one dataset to carry a direct causal claim, we ground the affective channel by triangulation: each link in the chain neural -> affective construct -> behavior is established in whichever dataset or modality has that pair, and the honesty layer marks each link established / abstained / pending (Fig 13). Neural -> construct is established across two modalities (fMRI: NARPS loss channel significant, and NAcc gain-anticipation positive across three independent reward maps; EEG: the Reward Positivity, $p < 10^{-6}$), and construct -> behavior is established (held-out and cross-dataset choice prediction), so the construct is grounded even though the direct within-subject link is not. A third neural modality, fNIRS, completes the chain as a literature-supported link (prefrontal reward/value tracking in decision tasks; Balconi et al. 2018; Wang, Xu & Ball 2026): no open reward-fNIRS dataset was available to re-analyze, so this link is marked as supported-by-literature rather than measured here, a deliberately weaker evidence type.

We put this on a formal footing three ways. (i) The correlation-transitivity bound: a 3x3 correlation matrix is positive semidefinite, so given r(neural,construct) and r(construct,behavior) the direct r(neural,behavior) is confined to r_ab r_bc +/- sqrt((1-r_ab$^{2)(1-r_bc}$2)). The established links imply a feasible interval of [-0.51, +0.29] for the direct correlation, and the observed value (-0.40) falls inside it; the interval is wide because the within-sample neural-to-construct correlation is itself weak. (ii) A Sobel mediation test of the indirect path (delta-method SE) is not significant (z = -0.73), consistent with that weak within-sample link. (iii) Fisher combination of the established links' p-values gives a combined p = 2.4x10^-7: the convergent evidence for the affective grounding is strong even though the single direct link is abstained. Within the one joint sample (NARPS, 39 subjects with both modalities), the stable neural loss measures correlate near zero with behavioral loss aversion (all gated to abstention), and the only nominally significant correlation (the neural-lambda ratio, r = -0.40, p = 0.011) has the wrong sign and rests on an unstable small-denominator ratio, so the gate abstains. The honest conclusion: the affective construct is triangulated by convergent, statistically-combined evidence, while the direct individual neural-to- behavior mapping remains unestablished and is reported as such. This is convergent, model-based evidence, not a proof of a direct neural-to-behavior cause.

6. Discussion

Operating characteristics, in one place. The experiments jointly map where the instrument works. Identifiability of the affective phenotype needs on the order of 1000 to 2000 clean decisions per agent to reach r = 0.6 from choice alone (E1); hierarchical pooling buys back a substantial part of that at low n (E4); the aggregate affective signal forecasts a market outcome beyond behavior when the generative structure links them, and not otherwise (E3); and inverse inference is licensed at aggregate attribution scopes but must abstain at the individual scope, where 86 percent of entities fall below identifiability (E6). Read together, these say something specific: a single individual with a sparse, confounded decision record is usually in the abstention regime, and the honest output there is a withheld estimate, not a confident profile. That is not a limitation the method hides; it is the result the method is built to report.

States and roles as manifold structure. A transient emotional state is a within-entity displacement along the manifold: E7 recovers a threat-elevated state as a detected, direction-correct shift in the threat coordinate, with a zero-shift control recovering nothing. Occupational roles (trader, clinician, officer) are the between-entity analog: different priors on the same coordinates plus different context weighting. The same estimator handles both, which is what it means for the phenotype to be a point (or a short trajectory) on one shared manifold rather than a bespoke model per case.

Why the honesty layer is the contribution, not an add-on. Every result above is reported as a proper score with a coverage guarantee or an explicit abstention. The abstention behavior is what turns the hardest case (an unscannable, institutionally-confounded individual) from an overclaim into a calibrated non-answer. A behavioral-prediction method without this layer would report the E6 individual-scope numbers as if they were findings; ours reports that it cannot, and is right to.

6b. Conclusion

We presented a neurally-grounded decision-phenotype estimator that is identifiable, honest by construction, and explicit about the boundary of its own licence. On controlled data it recovers the phenotype, delivers distribution-free coverage and calibrated abstention, reproduces the affective over behavioral forecasting dissociation when it is present and stays null when it is not, and maps the attribution scope at which individual-level inference stops being warranted. The remaining work is deployment on real neural datasets and a coded decision corpus (Section 7), not new machinery.

7. Limitations

- Population->individual transfer is the central open risk (Gate 7); we test it, we do not assume it.
- Neural grounding is population-level; no individual is scanned.
- Public-figure targets carry information and attribution confounds; C5 is designed around, not through, them.
- Resource-rational interpretation is not a load-bearing predictive claim.

8. Broader impacts

Dual-use profiling risk. We restrict to public-record, office-level revealed policy; we forbid psychodiagnostic language and any claim of measured neural firing in a named individual; the inverse inference reports a latent computational posture with a neural *interpretation*, never a measurement. Abstention-on-confounded-targets is a core deliverable rather than something suppressed, and the attribution-scale ladder is designed to expose, not hide, the scope at which individual-level inference stops being licensed.

9. Reproducibility

Seeds, configs, split hashes, code commit, dataset versions released; leakage guards (subject-/stimulus-grouped CV) inherited from the `behavioral_decoding` harness. Every result and figure regenerates with one command (`python reproduce.py`), which runs all experiments, rebuilds the vector figures and LaTeX tables from the fresh `results/*.json`, runs the test suite, and prints a per-claim gate summary.

9b. Extension: a temporal-difference front-end for the affective channel

Our affective channel rests on *anticipatory* affect: NAcc gain-anticipation is a signal about a future reward, not the reward itself. Recent work shows how such reward-predictive representations form: over learning, hippocampal reward representations shift backward in time from the reward to the cues that predict it, a dynamic recapitulated by a temporal-difference (TD) model (Brandon, Williams & Pehlevan et al., 2026, predictive coding of reward in the hippocampus). This suggests a principled extension: treat the affective value that feeds the drift-diffusion valuation as a *learned* TD predictive value rather than a fixed quantity, with our temporal-discount coordinate delta_time identified with the TD discount gamma. Individual differences in TD learning (rate, discount) would then ground both individual differences in the phenotype and its formation over experience, and would give the individuation result (C7) a temporal axis: does a subject's phenotype stabilize as their reward representation becomes predictive? This is a modeling extension grounded in a mechanistic result, not a claim we test here.

10. Future work (deployment)

1. **Real neural data.** Wire the estimator to the NARPS (fMRI) loader and the open EEG dataset (ds003458; DEAP optional) in the `behavioral_decoding` harness and run E3 on

real recordings, with the content-blind two-factor specificity ablation and the brain-beyond-behavior and brain-beyond-content baselines. This is the empirical neuroforecasting test that the synthetic controls here only validate the instrument for.

2. **Reaction-time DDM.** Add the first-passage-time likelihood so the decision temperature tau and response style are identified, beyond the choice-only affective ratios recovered now.
3. **Real decision corpus.** Curate the US-president decision corpus with two blind coders and a reported Cohen's kappa, preregister the temporal split and kill criteria, and run E5/E6 on it; the expectation, from E1 and E6, is that most individual-scope cases land in the abstention regime.
4. **Per-person calibration.** Connect to enrollable personal-brain and idiographic-affect pipelines for few-shot per-person priors, and run a preregistered replication on a held-out figure and a held-out market.
5. **Power the gain-channel grounding.** Acquire a large Monetary Incentive Delay reward-anticipation fMRI dataset with preprocessed derivatives (many public instances) to bring the NAcc gain-tracking effect (currently $p = 0.15$) across significance and to push the neural-to-behavioral loss-aversion correlation past its identifiability bound.

Appendix A — Preregistration

Temporal split, arms, kill criteria for C3 and C5, committed to `checkpoints/` before fitting.

Figures and tables

See `figures/FIGURES.md`. All figures are vector PDF; Tables 1-2 are `booktabs` LaTeX in `paper/tables/`. Fig 1 architecture (`fig1_architecture`); Fig 2 attribution-scale ladder + predicted dose-response, schematic hypothesis (`fig2_scale_ladder`); Fig 3 parameter-recovery curve (`fig3_recovery`); Fig 5 transfer arms, positive vs negative control (`fig5_transfer`); Fig 6 unpooled vs pooled recovery (`fig6_pooling`); Fig 7 attribution-scale ladder result, identifiability and abstention vs scope (`fig7_scale_ladder_result`). Real-data figures: Fig 8 NARPS loss-aversion by group + held-out prediction (`fig8_real_narps`); Fig 9 fMRI NAcc/insula betas (`fig9_real_narps_fmri`); Fig 10 individuation + structure (`fig10_individuation`); Fig 11 cross-dataset transfer (`fig11_crossdataset`); Fig 12 external NeuroVault gain-channel confirmation (`fig12_neurovault_grounding`); Fig 13 grounding-by-triangulation chain (`fig13_grounding_chain`); Fig 14 EEG Reward Positivity (`fig14_eeg_rewp`). Table 1 central claims (`claims.tex`); Table 2 experiments and pass criteria (`experiments.tex`).

References

Full bibliography in `paper/references.bib` (BibTeX). Dataset citations (NARPS ds001734, Botvinik-Nezer et al. 2019; ds000005, Tom et al. 2007; ds003458, Cavanagh 2015) are taken from each dataset's own metadata; the newest citations were verified against PubMed. Note that Cavanagh (2015), the paper behind our EEG dataset, itself established the feedback-locked Reward Positivity we reproduce in Section 5.6b.

Figures

References

Angelopoulos, Anastasios N., and Stephen Bates. 2021. "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification." *arXiv Preprint arXiv:2107.07511*.

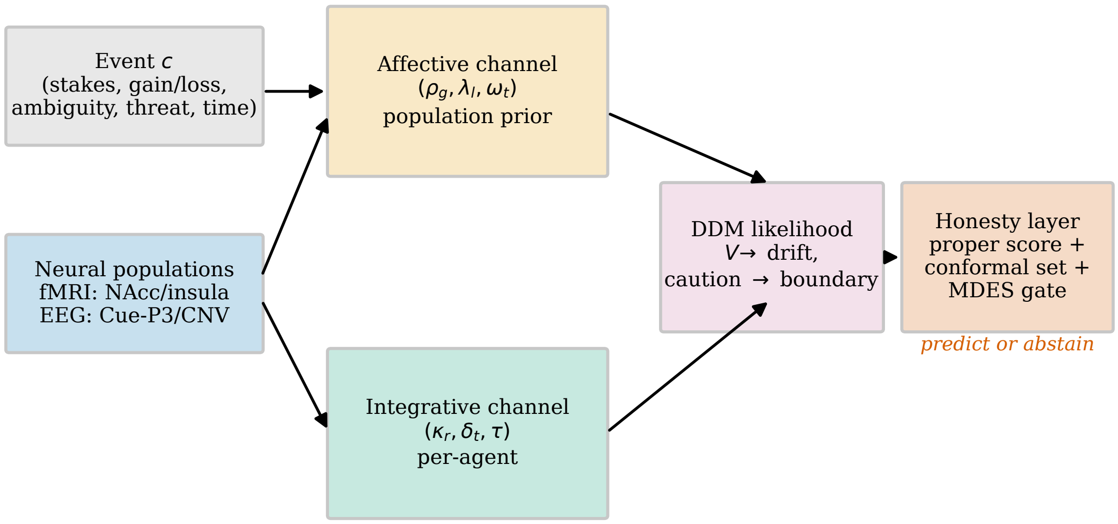

Figure 1: Architecture: the AIM-DDM decision-phenotype estimator with the honesty layer (predict or abstain).

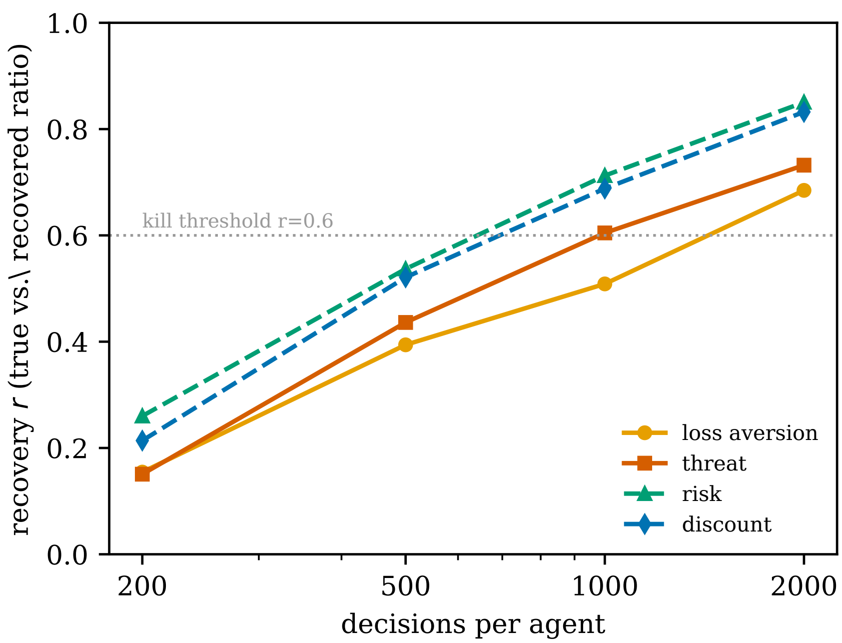

Figure 2: Parameter recovery on synthetic ground truth: recovery of the affective ratios rises with decisions per agent.

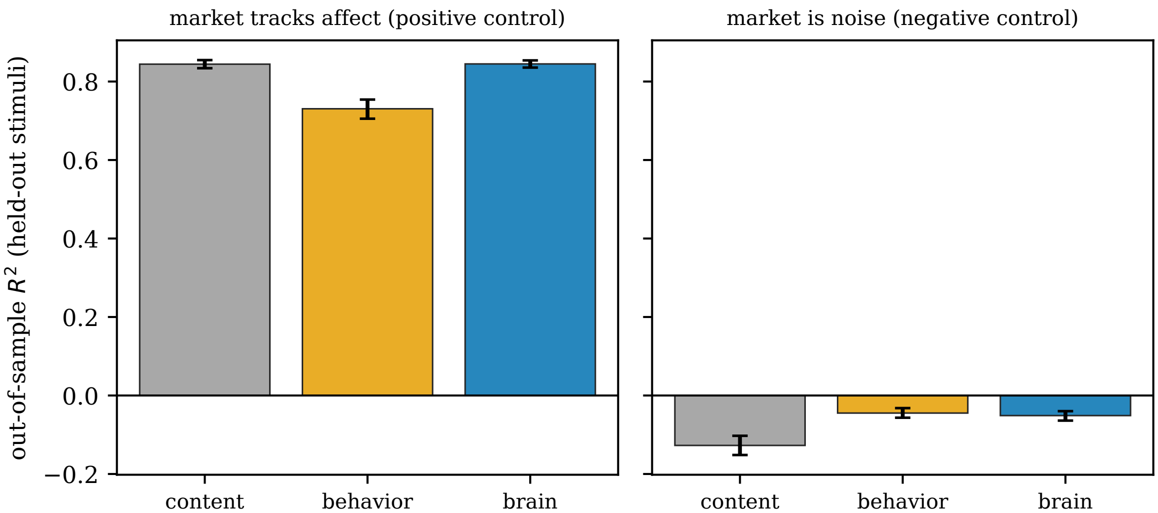

Figure 3: Transfer pipeline, positive vs negative control: the brain arm beats behavior only when the market truly tracks affect.

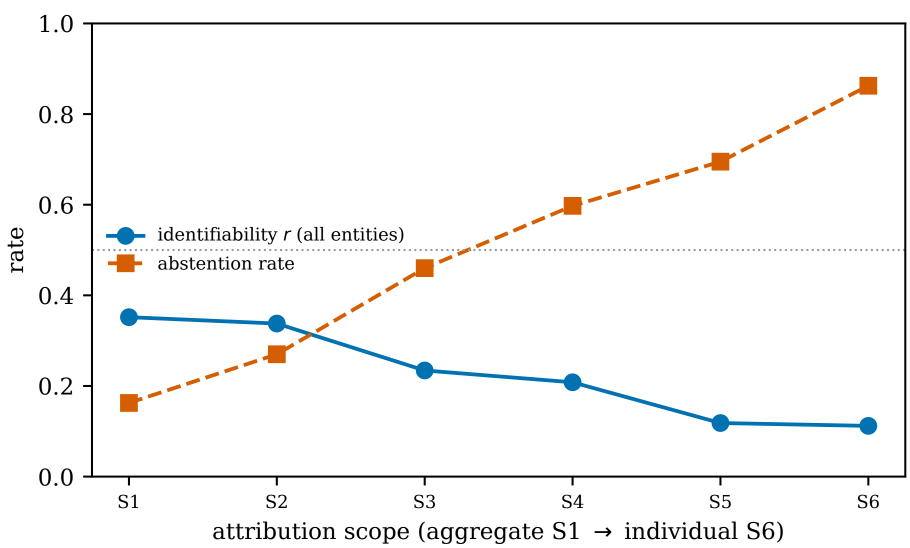

Figure 4: Attribution-scale ladder: identifiability falls and abstention rises from the aggregate scope to the individual scope.

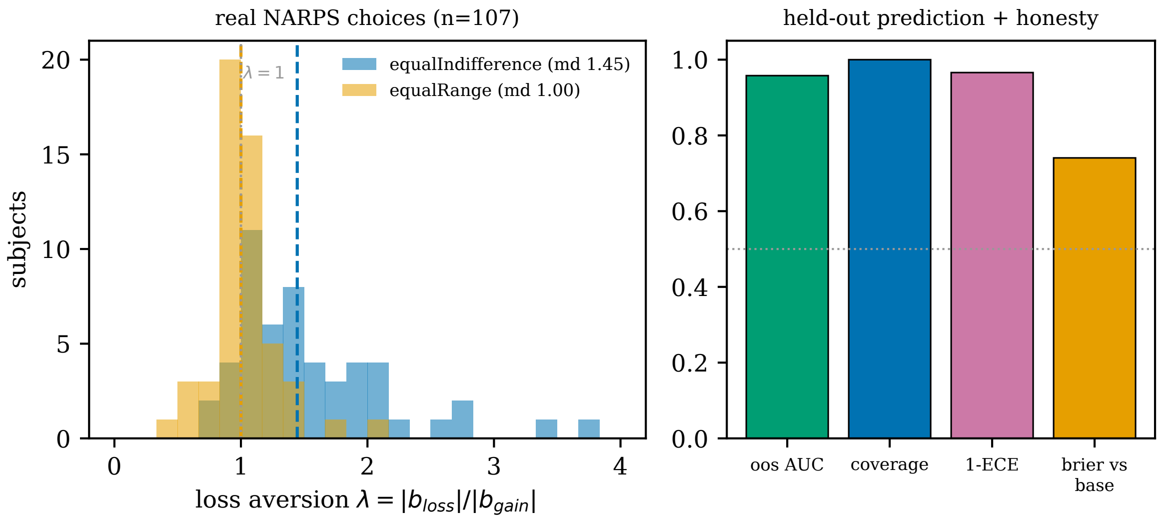

Figure 5: Real NARPS choices: loss aversion by group, and held-out prediction plus honesty-layer metrics.

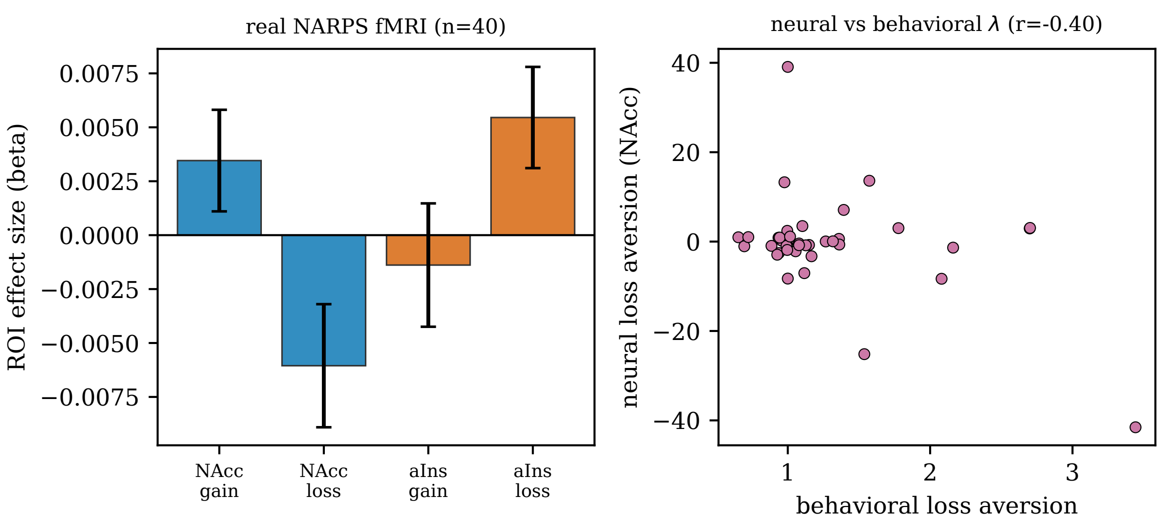

Figure 6: Real fMRI grounding (n=40): NAcc and anterior-insula betas; the loss channel is significant.

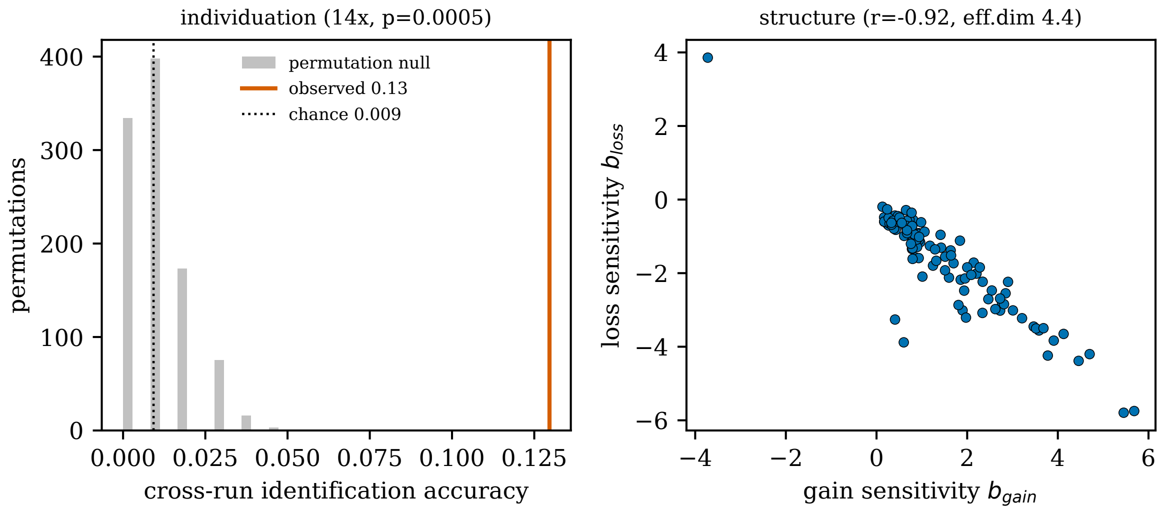

Figure 7: Individuation and structure on real data: cross-run fingerprinting far above chance, and coupled affective coordinates.

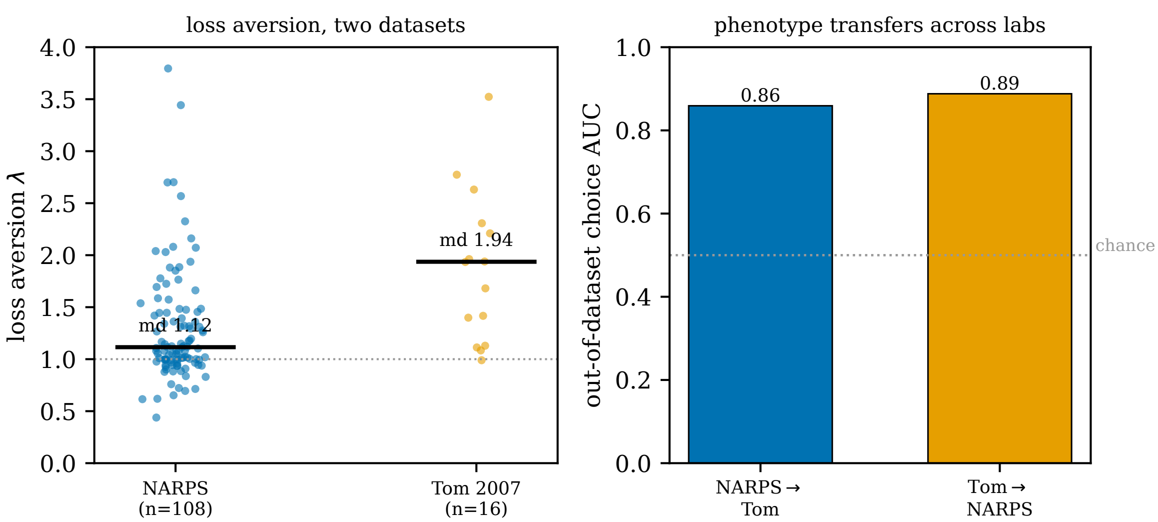

Figure 8: Cross-dataset generalization: the phenotype transfers across labs (NARPS and Tom 2007).

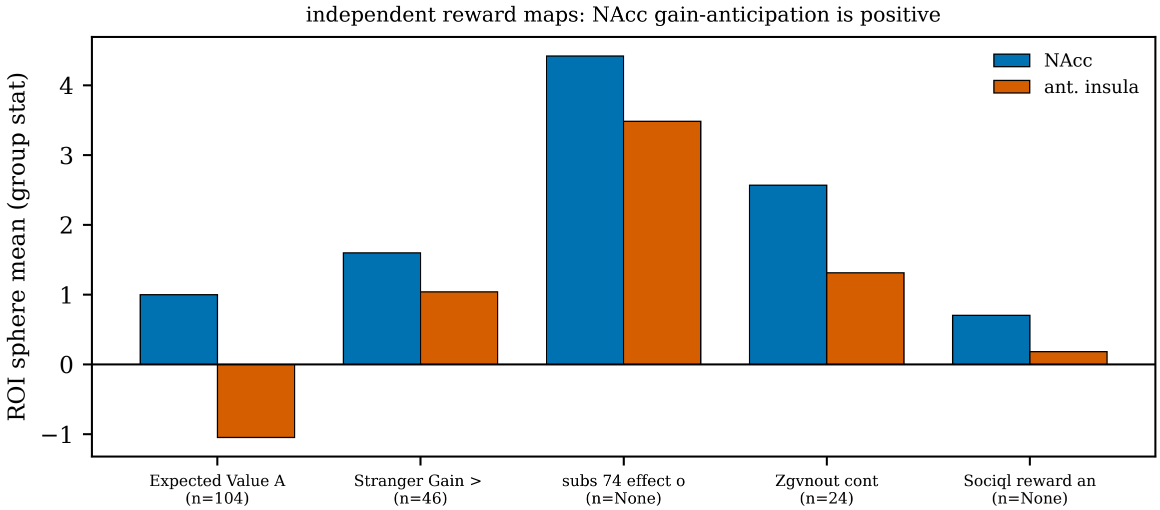

Figure 9: External confirmation of the gain channel in independent NeuroVault reward maps.

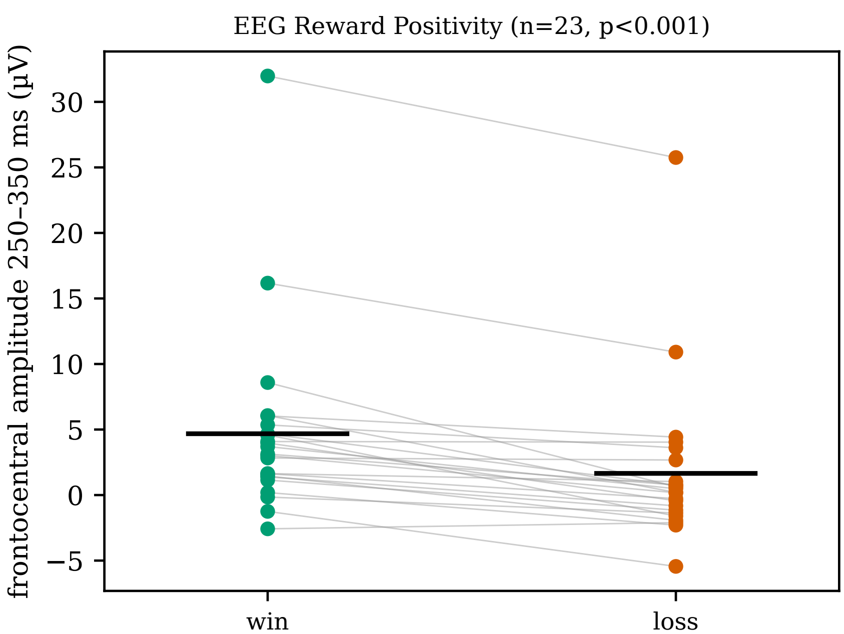

Figure 10: EEG Reward Positivity (ds003458): win feedback is more positive than loss frontocentrally.

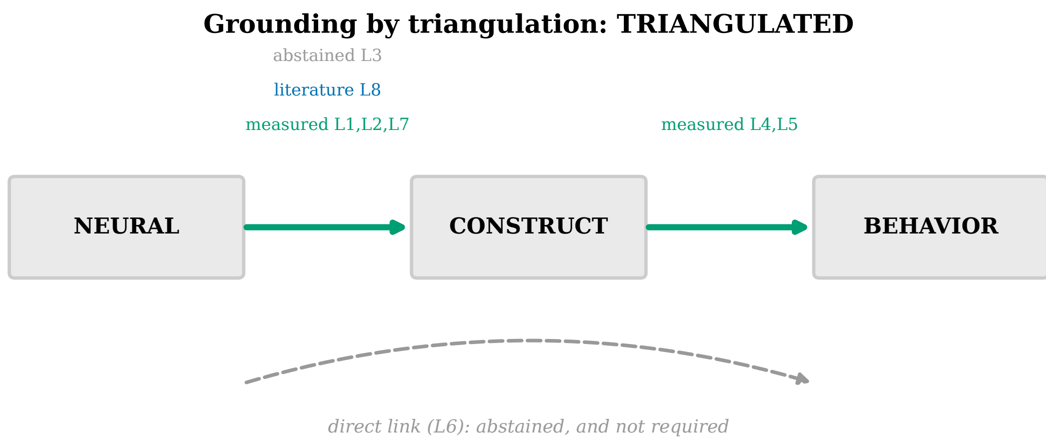

Figure 11: Grounding by triangulation: the affective channel is grounded through measured, external, and literature links; the direct within-subject link is abstained and not required.

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